Augmented Reality Interface for Direct Robotic Welding Path Programming

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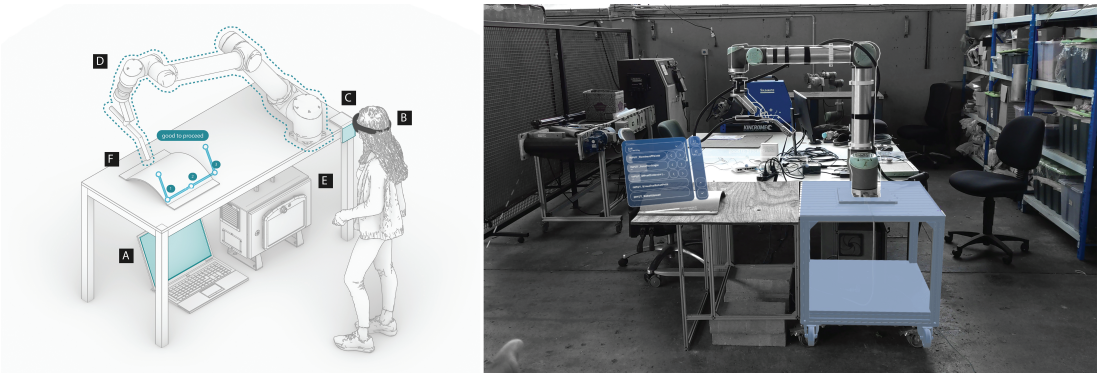


Figure 1: [Left] Proposed AR-assisted robotic welding setup. (A) Workstation, (B) Microsoft HoloLens 2, (C) Visual marker, (D) UR10, collaborative robot, and (E) Control box. [Right] Successful tack welding path preview within the proposed AR interface, illustrating trajectory validation.

Abstract

Programming robotic welding tasks remains challenging for non-experts due to the complexity of traditional teach-pendant and CAD-based methods. Augmented Reality (AR) presents a promising alternative by enabling intuitive, spatially contextualised interactions directly within the physical workspace. Prior work related to robotic welding programming has primarily focused on seam welding through Programming-through-Demonstration (PtD), with limited attention to tack welding tasks. We present a proof-of-concept AR interface that supports interactive definition for both seam and tack welding paths using gesture-based spatial input on a physical workpiece. Our system also incorporates real-time reachability feedback and supports a human-in-the-loop (HITL) planning workflow. An informal evaluation with research team members suggests feasibility, though challenges such as spatial anchoring drift remain. We contribute: (1) an AR-based robotic path planning workflow that supports both tack and seam welding, and (2) an interactive interface that leverage leveraging real-time spatial feedback and natural gestural input to support users with limited robotics experience.

Index Terms: Augmented Reality (AR), Head-Mounted Display (HMD), Human-Robot Interaction (HRI), Robotic Welding, System Design.

1 Introduction and Related Work

Welding is an essential, yet physically demanding, and hazardous profession that requires significant precision and expertise [3]. Despite its critical role in manufacturing and construction settings, industries face a steady decline in skilled welders, driven largely by occupational health risks and physical strain [1]. In recent years, robotic welding systems have been proposed as viable solutions to improve worker safety, productivity, and consistency [4]. However, the adoption of robotic welding remains limited. This challenge can be broadly divided into two core problems: path planning and execution. Given the inherent complexity of robotic welding tasks, this research focuses exclusively on robotic path planning. Conventional robotic programming methods for welding tasks include the use of teach pendant or CAD-based programming, both of which are often time-consuming and require accurate representation of the virtual environment [6].

Augmented reality (AR) has recently emerged as a promising solution to overcome these limitations, as it simplifies robotic welding programming through intuitive, spatially contextualised interfaces [5]. AR has been deployed alongside Programming-through-Demonstration (PtD) techniques for robotic seam welding. As demonstrated in [5], the authors use this approach to support non-expert users to define welding paths. It is important to note that the accuracy of PtD heavily depends on the quality of the demonstration and the capability of the motion capture tool. Recognising this limitation, [4] have explored how path optimisation algorithm can be used to improve the precision of the demonstrations.

Existing research on AR-supported robotic welding is primarily associated with the PtD paradigm. However, one of the shortcomings of the PtD approach in robotic welding is its inability to automatically segment the weld seam for tack welding. Thus, tack welding remains relatively underexplored despite its significant role in manufacturing settings. Our work addresses this gap by leveraging computational capabilities and AR to assist users in specifying both the number and placement of tack points along the seam.

In this work, we present a proof-of-concept AR-based system that allows users to intuitively define both tack and seam welding paths without prior robotics knowledge. To the best of our knowledge, this study is the first to specifically address AR-assisted tack welding, thereby expanding the scope of AR-supported human-robot interaction (HRI) research into a crucial yet underexplored area.

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2 System Design

2.1 System Architecture: Hardware

The hardware system comprises three main components: a see-through head-mounted display (HMD), a cobot, and a Windows PC (illustrated in Figure 1). Microsoft HoloLens 2 was used to enable immersive hands-free visualisation and facilitate spatial interaction with weld paths through the Fologram application [2]. The robotic workcell includes an UR10 collaborative robot (cobot) that executes the welding path defined by the users.

The workstation and cobot are connected via an Ethernet cable to ensure reliable transmission of robot commands. At the same time, the HMD communicates with the workstation over a stable local Wi-Fi network. The AR visual environment is spatially anchored on the cobot's physical workspace via a visual marker.

2.2 System Architecture: Software

The system is built on *Grasshopper*, a computationally driven programming platform embedded in Rhino 3D. Weld points and seam paths are generated using geometry-based definitions, and robot trajectories are planned through the Robots plugin, which supports the simulation and export of URScript-compatible code.

Fologram [2] bridges the AR and programming environments by streaming the Grasshopper model to the *HoloLens 2* in real time, allowing users to view and interact with weld plans superimposed on the physical workpiece. All parameter changes in the AR environment are synchronised with the underlying Grasshopper model.

2.3 Interaction Workflow

Users interact with the system through an AR interface rendered on the *Microsoft HoloLens 2*. A virtual spatial menu allows users to switch between tack and seam welding modes. Within each mode, users can adjust design parameters such as (A) welding speed, (B) welding angle, (C) offset from the metal workpieces, and (D) resolution (number of target points). In addition, users can specify the number of tack welds as well as the locations while in tack welding mode.

Weld paths are defined by manipulating virtual widgets using pinch-and-drag gestures. These widgets are directly associated with robot poses and tool orientations. As users adjust cobot waypoint(s), a simplified trajectory preview is rendered within the visual environment, providing real-time visual feedback on the cobot's reachability. The AR environment also incorporates colour-encoded visual cues and text annotation (refer to Figure 2) to indicate reachability constraints. These elements help users identify and correct infeasible paths prior to execution. Once validated, users transmit the movement command directly to the cobot, completing the planning execution loop.

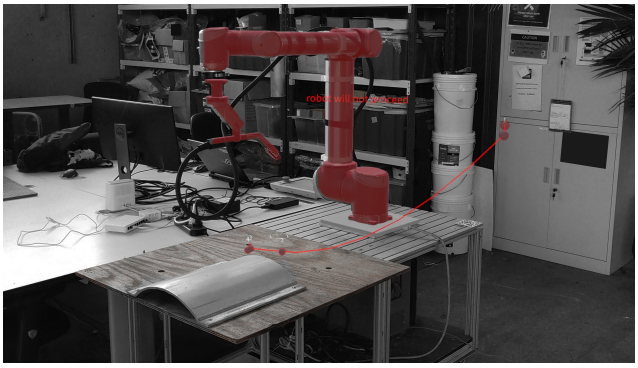


Figure 2: Screenshot of AR interface preview highlighting unreachable weld paths and robot meshes in red with text annotations indicating reachability constraints

3 Results and Discussion

Given the exploratory nature of this proof of concept, we performed an informal evaluation with three participants (N = 3) that included an author and two colleagues to assess preliminary feasibility rather than full usability or performance results. Each participant completed the setup of the seam and tack welding task in approximately 5-10 minutes without prior training. Each participant completed the path planning for both seam and tack welding in approximately 5-10 minutes without prior training. Spatial anchoring remained relatively stable, but the virtual environment drifted noticeably during longer trails or when the visual marker was not within the field of view of the Microsoft HoloLens 2. The participants found that AR-based gesture interactions were intuitive, although occasional precision errors occurred when adjusting robot waypoints.

These early findings suggest the system's potential for streamlining robotic welding setup, particularly in environments where rapid prototyping is critical. Although geometry-based interactions effectively supported tack welding, seam welding may benefit more from traditional AR-supported PtD as shown in [4, 5]. Given that welding practice involves significant tacit knowledge, PtD could capture nuances that our proposed approach may overlook. However, features such as real-time reachability feedback remain valuable for users with limited robotics expertise.

4 Conclusion

This poster presented a proof-of-concept system that integrates an AR-based interaction for tack and seam welding tasks. By allowing users to define weld paths directly within a spatially anchored AR environment using gestural inputs, the system offers a more simplified and flexible approach to robot programming.

As a proof-of-concept, our evaluation is preliminary and informal; future work will involve more structured usability studies to evaluate task efficiency, learning curves, and cognitive load. Moreover, the research could also adopt a more user-centered design approach by involving skilled welders in iterative system development to ensure that the developed system meets their needs.

Supplemental Materials

Supplementary materials will be available after the review process.

Acknowledgements

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